

Week 8 · Laboratory Problems — build the position controller in Simulink

- Course: Sensor Signal Processing and Fusion · Department of Autonomous Vehicle System Engineering
- Time: **one hour**, immediately after the Week 8 lecture hour
- Three problems, in order.

What this hour is for

Week 6 §6-3 established that the sway row of **B** is zero: this hull cannot push sideways. This hour is what that costs, and what is done about it.

Problem 1 builds a position controller that is **correct** and watches it fail — the force it asks for is right, the whole time, and none of it can be delivered. Problem 2 changes one line and the same vessel holds station in the same current. Problem 3 gives it three waypoints and meets the failure that only appears when a working controller is put inside a mission.

Before starting

```
cd lectures/w08_simulink/problems
w08_P1_start           % creates W08_P1.slx — the mission, the hull, nothing between
w08_check(1)           % run this whenever, as often as needed
```

The solver is already set to fixed-step `ode4` at `h = 0.02` s. **Do not change it.**

★ One requirement on every model

`dlog` — To Workspace, `Structure With Time`, carrying $[\psi_d; X; N]$: the heading **the controller commands**, in degrees, and the two demands it makes, in newtons and newton-metres.

ψ_d is logged because Problem 2 is about a heading the controller **chooses**, and a vessel can end up pointing into a current for reasons that have nothing to do with the rule under test. The checker compares the command with the heading actually reached, which no position measurement could do.

The mission logic, Week 6's allocation and the hull are provided. `mlog` carries $[N_d; E_d; \text{mode}]$ — what the mission asked for — and `xlog` the twelve states.

Problem 1 · The direction it cannot hold (20 minutes)

Build. A PID on the position error, **in both north and east**, and then the only part of it that can reach the propellers.

$$f_N = K_{p,x}e_N + I_N - K_{d,x}v_N, \quad f_E = K_{p,x}e_E + I_E - K_{d,x}v_E$$

$$X = f_N \cos \psi + f_E \sin \psi, \quad \psi_d = \psi_{\text{fix}}$$

Symbol	Quantity	Value · source
$K_{p,x}, K_{i,x}, K_{d,x}$	the position gains	30 N/m, 3, 60 · §8-4
v_N, v_E	the velocity in NED	m/s · $u \cos \psi - v \sin \psi$ and $u \sin \psi + v \cos \psi$
ψ_{fix}	the heading this problem holds	0°, due north
V_c, β_c	the current	0.3 m/s towards 90° · Week 1 §1-13

The integral needs the anti-windup of Week 2 §2-11: accumulate only while $|X|$ is inside its limit.

i
 X is a projection, not a choice

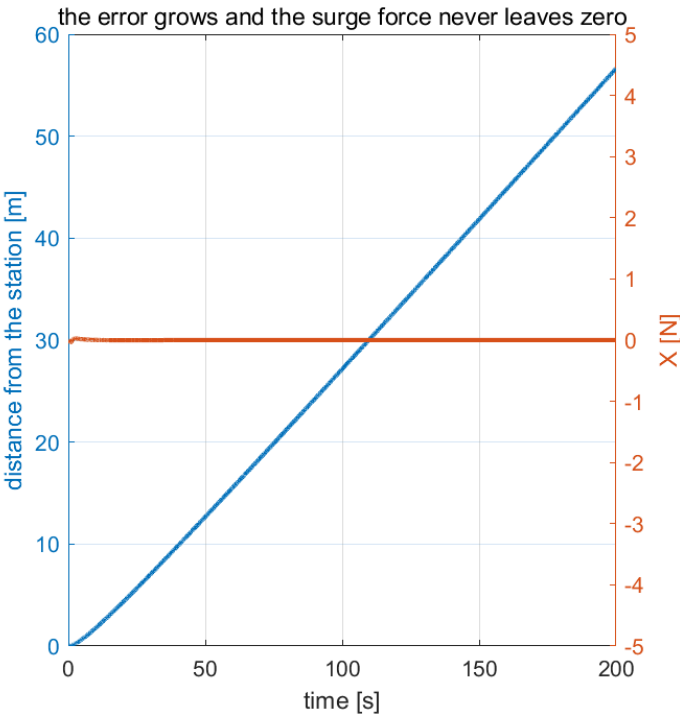
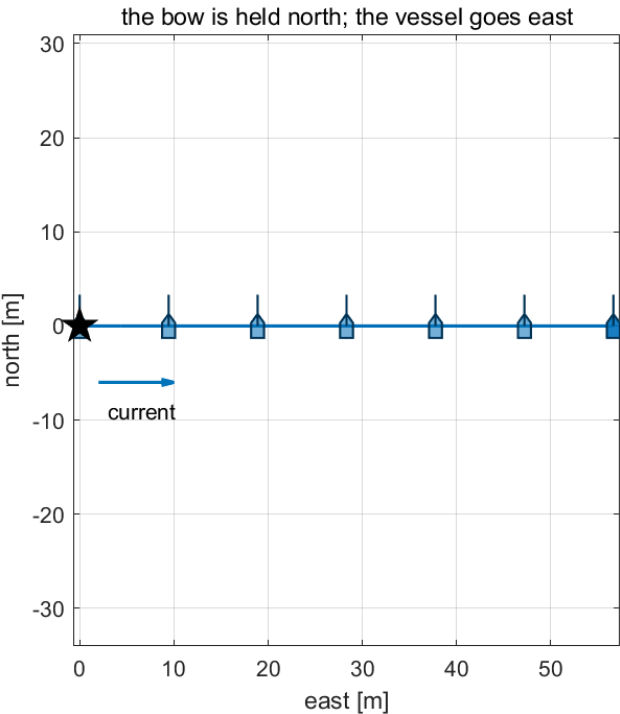
The position loop asks for a force in a direction. The propellers can produce force in **one** direction — along the bow. What reaches them is therefore **f** projected onto the bow, and the component across it is not weakened or delayed: it is unavailable.

Predict before running. The current runs east; the vessel is asked to hold a station; the bow is held north. Work out which way **f** will point once the integral has settled, and then what its projection on a northward bow will be.

Verify. `w08_check(1)` .

What the checker expects, after 200 s	
distance from the station	56.64 m
the largest surge force ever asked of it	0 N
the heading	never moves

What a correct model produces



Reading the figure	
left	the hulls all point north while the track runs east
right, the error	rising without limit; nothing arrests it
right, the force	flat on zero for the whole run
the check	if $\mathbf{X}$ is large and the vessel still drifts, the projection has been written with sin and cos exchanged

The point. Nothing here is a fault of the controller, and no gain will repair it. The loop asks for a force to the west, correctly and continuously; its projection on a northward bow is zero; and the sway row of **B** is zero, so no pair of shaft speeds can produce what is missing. **This is underactuation, not bad tuning** — and the failure looks exactly like bad tuning, which is why it is worth building.

Problem 2 · Point the bow along the force (20 minutes)

Build. One line, replacing $\psi_d = \psi_{\text{fix}}$:

$$\psi_d = \text{atan2}(f_E, f_N) \quad \text{while } \|\mathbf{f}\| > e_{\min}$$

and then Week 4's autopilot, **gains unchanged**:

$$N = K_p \text{ssa}(\psi_d - \psi) - K_d r, \quad |N| \leq N_{\max}$$


Hold the last heading when the force is small

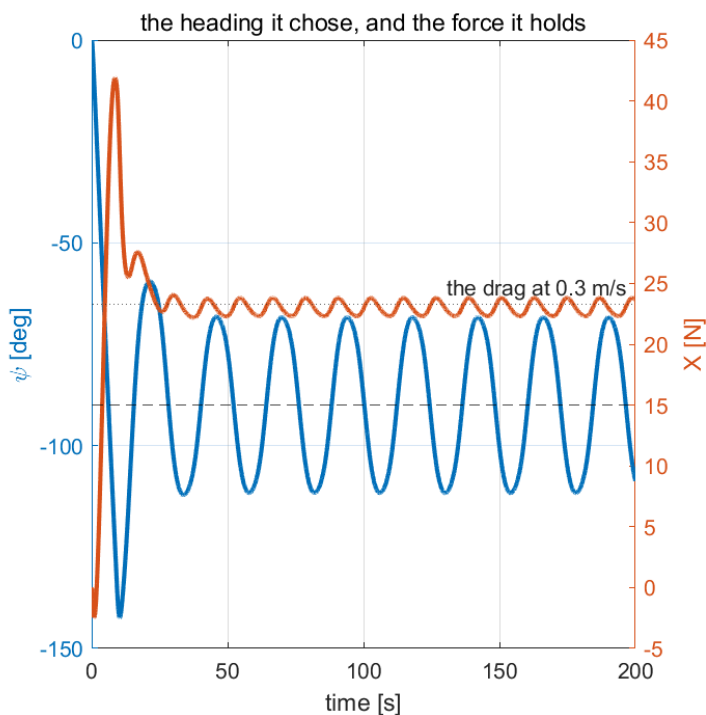
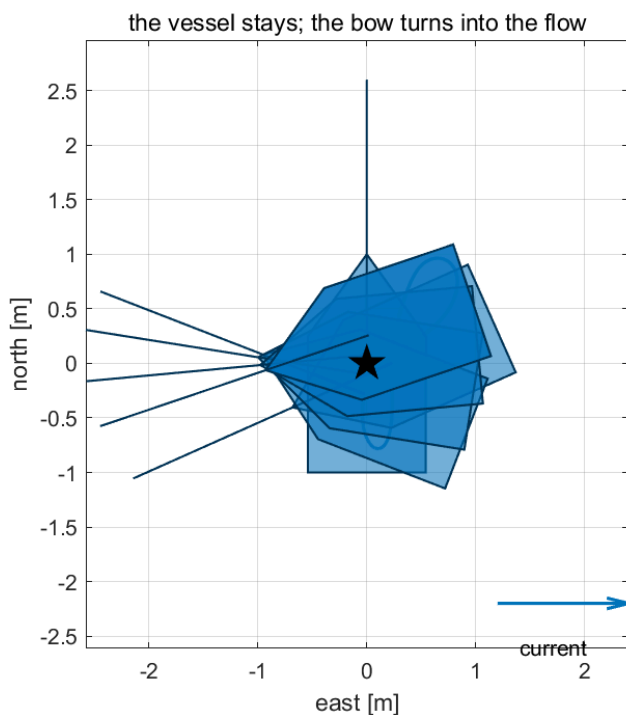
Near the station **f** is almost zero and its direction means nothing: **atan2** of two numbers that are both noise is noise. Below $e_{\min} = 0.3$ N, keep the heading the rule last produced.

Predict before running. The integral will settle on a force that exactly opposes the current. Say, before running, which way the bow will end up pointing and how large $\mathbf{X}$ will be — the drag of this hull at 0.3 m/s is 0.3/0.012894 N.

Verify. `w08_check(2)`, over the last 50 s.

What the checker expects	
distance from the station	0.193 m
the heading it settles at	−90.6°
the surge force it holds	23.0 N, against a drag of 23.3 N
the heading error	0 — the bow really did follow the command

What a correct model produces



Reading the figure	
left	the vessel stays on the star while the hulls swing round to face the flow
right, ψ	settling onto the dashed -90° , which is into a current running towards $+90^\circ$
right, X	settling onto the dotted drag line
the check	if the bow spins slowly and never settles, $e_{\min}$ is too small and the rule is steering by a force that is almost zero

The point. The same current, the same gains, and the station is held. **Nothing was added to the position loop.** The integral learned an upstream force and the bow was allowed to follow it — and that the force it settles on equals the drag at **0.3 m/s** is the check that the vessel is doing physics rather than merely holding still.

This is weathervaning, and it buys position by giving up heading. A mission that needs both would need a different hull, and §6-3 says exactly which row of **B** would have to stop being zero.

Problem 3 · The mission, and the handover (20 minutes)

Build. Two conditions, in the controller of Problem 2.

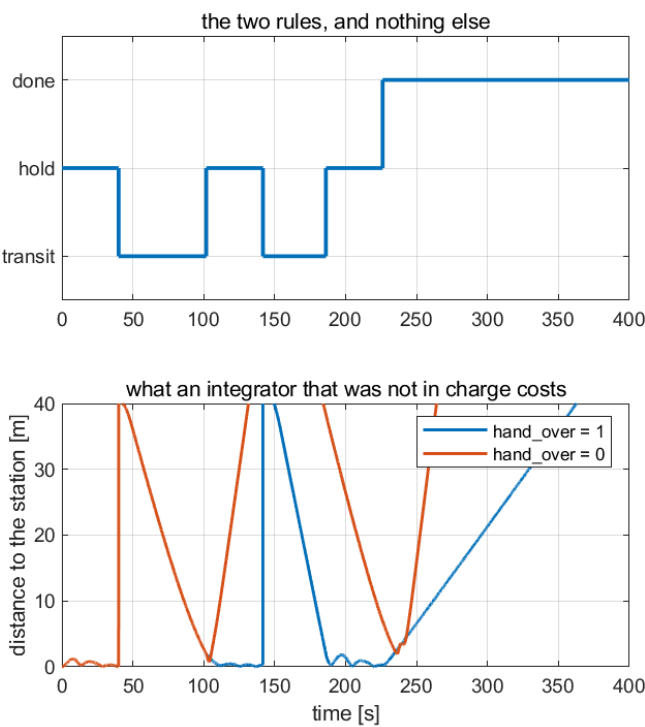
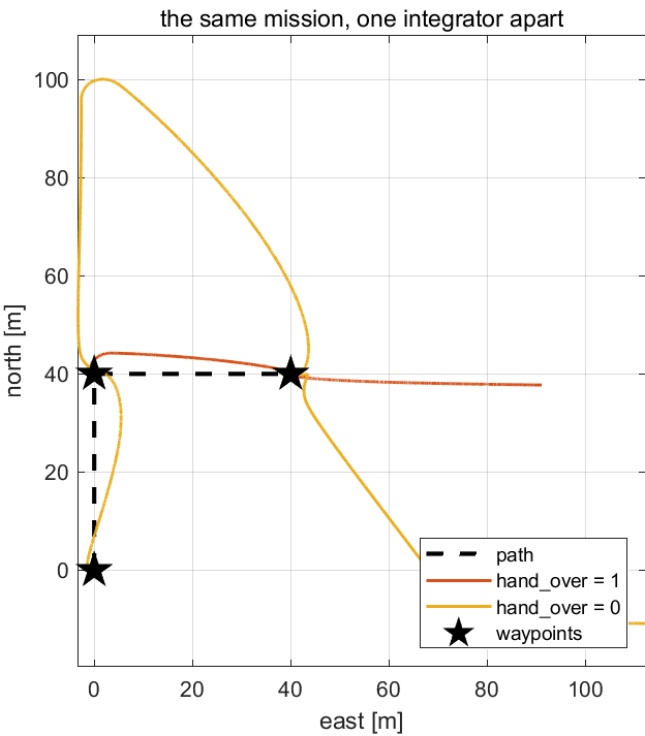
- In transit (`mode == 1`): push with a constant $X_{\text{ff}} = 60$ N and aim the bow at the waypoint, $\psi_d = \text{atan2}(e_E, e_N)$.
- At a change of mode, **empty the position integrals** — that is `hand_over`.

Predict before running. During a transit the position error is tens of metres and the integral is accumulating all of it. Work out what I_N would demand at the instant a hold begins, and compare it with $X_{\text{max}} = 120$ N.

Verify. `w08_check(3)`. Three waypoints, 40 s at each, flown twice.

What the checker expects	
the modes	five changes, alternating, ending in <code>done</code>
each hold	exactly <code>40</code> s
<code>hand_over = 1</code>	peak $ X $ of <code>43.3</code> N in the first <code>10</code> s of the second hold; that hold keeps <code>0.420</code> m
<code>hand_over = 0</code>	<code>120.0</code> N — the limit — and <code>25.626</code> m

What a correct model produces



Reading the figure	
left, the orange track	straight runs between knots at the waypoints
left, the yellow track	the same controller, thrown <code>60</code> m past the first station and <code>70</code> m past the second
top right	five changes, alternating, and then <code>done</code> : the two rules and nothing else
bottom right	the error falling into each hold and rising in each transit, on one trace only
the check	if the modes never leave <code>hold</code> , the transit branch is not reading <code>mode</code>

The point. The integrator is the one state that does not respect modes. Everything else in the controller is a function of the present error; the integral is a memory, and when the mode changes it is a memory of a situation that has ended. It demands what it stored — the limit — and throws the vessel past the station before it unwinds.

A controller that is not in charge must not integrate. **Integration is where a controller that works and a mission that works stop being the same thing.**

Marking

	Weight	What is being marked
Problem 1	30	<code>W08_check(1)</code> passes; the sentence "this is underactuation, not bad tuning" is argued from the rank of B
Problem 2	35	<code>W08_check(2)</code> passes; the settling heading and the drag force were predicted before running
Problem 3	35	<code>W08_check(3)</code> passes; the size of the demand a carried-over integral makes was worked out before it was measured

If something goes wrong

Symptom	Cause	Fix
The model has no To Workspace block whose variable name is <code>dlog</code>	only the provided logs are there	add the third: three signals, <code>Structure With Time</code>
The vessel drifts away with X large	sin and cos exchanged in the projection	$\mathbf{X} = \mathbf{f}_N \cos \psi + \mathbf{f}_E \sin \psi$
The vessel drifts away with X at zero	that is Problem 1, and it is the correct answer	set the bow free — Problem 2
The bow wanders slowly on station	$e_{\min}$ too small: steering by a force that is nearly zero	raise it, or hold the last heading below it
The bow points downstream	the arguments of <code>atan2</code> are exchanged	$\psi_d = \text{atan2}(\mathbf{f}_E, \mathbf{f}_N)$, east first
The vessel turns continually and never settles	the position loop is outrunning the heading loop	that is §8-4's cascade rule; lower $K_{p,x}$
The mission never leaves the first waypoint	the transit branch is not reading <code>mode</code>	it is an input to the controller, not a parameter
Every hold begins with a violent lurch	<code>hand_over = 0</code>	that is Problem 3, and it is the point of it

Reference answers are in `../solutions/`. Read them **after** attempting the problem.